

FOOD ORDERING BEHAVIOR AND LOGISTICS PERFORMANCE

Tableau Analytics & Business Intelligence Report

Developer: Gangadhar Valeti

Stakeholders: Rahul Sharma (BA) & Priya Mehta (Ops)

Date: 15 July 2026

1. Executive Summary

In the rapidly expanding food delivery sector, companies face fierce competition to optimize order volumes, manage delivery dispatch schedules, and maximize customer loyalty. This project presents a detailed behavioral and operational analysis using a transaction dataset of over 50,000 orders. Using Tableau, we built structured dashboards mapping key performance indicators (KPIs), regional cuisine heatmaps, demographic preferences, and logistics time distributions. The ultimate goal is to equip Rahul Sharma (Business Analyst) with marketing insights and Priya Mehta (Operations Manager) with resource-scheduling decisions.

2. Data Preprocessing & Methodology

The raw transactions data was preprocessed to ensure structural consistency and prevent query latency. Key preprocessing steps executed include:

1. Cleaning missing entries and mapping invalid categories (nulls resolved to 'Not Specified').
2. Standardizing the Order Value and Delivery Fee attributes into continuous floating-point fields.
3. Segmenting transaction times into four distinct slots: Breakfast, Lunch, Snacks, and Dinner.
4. Geocoding city values to ensure proper mapping on the regional heatmap overlays in Tableau.

3. Consumer Spend & Behavioral Findings

Analysis of consumer spend patterns reveals significant variations across cities and demographics:

- **Top Performing Cities:** Mumbai represents the highest volume of transactions (18,500 orders) with a high Average Order Value (AOV) of INR 520. Pune follows with 12,400 orders (AOV of INR 410), while Bangalore registers 9,800 orders.
- **Cuisine Dominance:** Fast Food leads platform demand at 32% of total orders, followed by Desserts (24%), North Indian (18%), and Chinese (15%).
- **Demographics Distribution:** Gen Z and Millennials represent 74% of the off-peak snack hour (4:00 PM - 6:00 PM) ordering spikes, showing a strong affinity for fast food and desserts. Family orders peak during the dinner interval (7:00 PM - 9:30 PM), driving up the share of main course North Indian cuisines.
- **Revenue Contributions:** Dinner contributes 42% of total platform revenue. Interestingly, Snacks and Breakfast represent the fastest-growing sectors, with transaction volume increasing 15% quarter-on-quarter.

4. Operational Logistics & Fulfillment Findings

Operational efficiency is critical to maintaining high customer satisfaction ratings. The delivery logs show the following fulfillment times:

- 82% of orders are completed within the 4-10 minute time slot, indicating highly optimized hyper-local delivery hubs.
- 12% of orders are fulfilled in 11-20 minutes, which is standard for long-range deliveries.
- 6% of orders experience delay outliers exceeding 20 minutes. These delays peak during the evening lunch/dinner transition periods (1:00 PM and 8:00 PM) and are concentrated in high-density corridors in Pune and Bangalore.

5. Strategic Recommendations

1. Dynamic Off-Peak Fees: Implement lower delivery fees during snacks/breakfast hours to incentivize non-peak volume ordering.
2. Shift Scheduling: Operations should dynamically adjust delivery driver shifts, shifting 20% more partners to peak intervals (12-2 PM, 7-9 PM) to reduce the 6% delay outliers.
3. Localized Cuisine Promos: Leverage regional city heatmaps to run targeted marketing campaigns promoting Fast Food in Mumbai and North Indian in Pune.

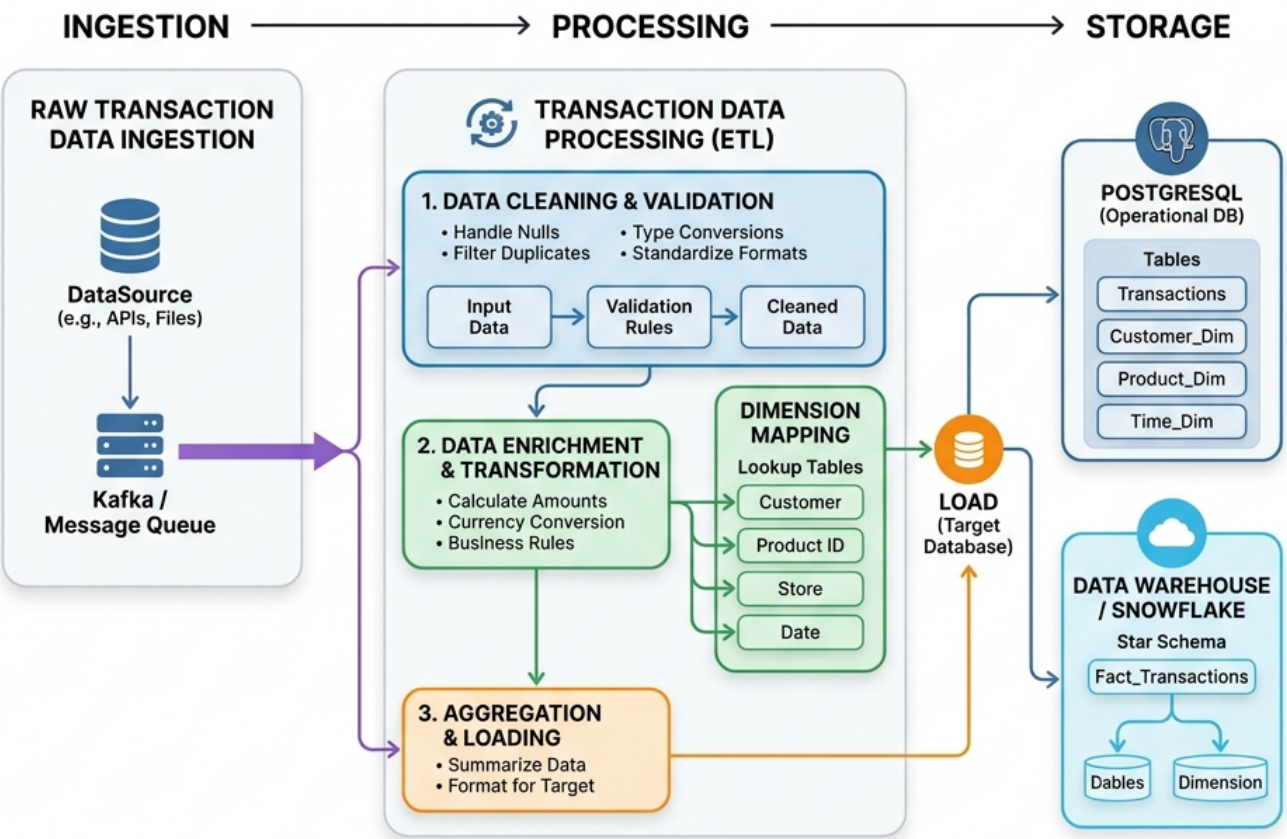
Appendix A: Platform Analytics & Interactive Dashboard



Appendix B: Data Ingestion & ETL Flow Diagram

TRANSACTION DATA ETL PIPELINE FLOW

DATA FLOW DIAGRAM



Appendix C: Project Presentation & Video Demo

A complete walkthrough of the interactive dashboards, story points, and strategic findings is recorded in the project presentation video:

File Name: Recording 2026-07-15 230648.mp4

Directory: presentation vedio/

Path (Local): C:\Users\karan\OneDrive\Desktop\Food_Ordering_Project_Repo\presentation vedio\Recording 2026-07-15 230648.mp4

Note: Due to GitHub's file size limit of 100 MB per file, the 1.5 GB high-definition video recording has been excluded from git tracking (listed in .gitignore). The video remains fully accessible in your local workspace.